

Thilak Kumar Akula

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EDUCATION

Catholic University of America, Washington Dc

Master of Science in Data Analytics (CGPA: 3.05/4)

August 2023-May 2025

Courses: Data Visualization with Tableau, Database Systems, Python for data science, Machine Learning for Data Science

CMR University, Bangalore, India.

May 2023

Bachelor of Technology in Cloud Technology and Information Security (CGPA: 9.34/10)

Courses: Machine Learning, Data Warehousing and Mining, AWS, Operating Systems, Data Structures, DBMS

TECHNICAL SKILLS

- Programming languages: Python.
- Libraries: Scikit-learn, NumPy, Pandas, NLTK, TensorFlow, PyTorch
- Web Technologies: HTML, CSS, Bootstrap, JavaScript, jQuery, JSON, Django, Flask, React.js
- Tools: PyCharm, Jupyter notebook, Anaconda, Visual Studio, Git, Power BI, Terraform, Docker.
- AWS Services: CloudFormation, S3, Lambda, Fargate, SNS, SQS, DynamoDB
- Databases: MySQL, PostgreSQL, MongoDB
- Operating Systems: Windows, Unix/Linux, Android

EXPERIENCE

Generative AI, Presto, Bengaluru, India

August 2023-July 2024

- Collaborated with the AI team to design and test algorithms for voice automation solutions.
- Analyzed large datasets to train and fine-tune AI models, ensuring high accuracy.
- Contributed to the development of AI-driven features for Presto voice.
- Technologies: MySQL, NumPy, Pytorch, TensorFlow, Pandas, Python.

Prompt Engineer -NLP/Python, Delta cubes, Bengaluru, India.

January 2023-

March2023

- Evaluated and improved prompt effectiveness using user feedback and iterative design.
- Identified key details from audio content to create precise prompts and summaries.
- Ensured prompt requirements were clear and unambiguous for accurate responses.

PROJECTS

Advanced Prompt Engineering for Customer Support Automation

May 2024

- Developed a prompt engineering framework to enhance a GPT-4 modal for automating customer support.
- Improved response accuracy by 25% and reduced response time by 40% Integrated the solution into existing platform and achieved a 30% increase in user engagement.

Cloud-Based Machine Learning Model

May 2023

- Implemented an android application that provides an extractive summary of an article or text document with an aim to save people's time and efforts.
- Performed text preprocessing using NLTK library and trained Restricted Boltzmann Machine neural network to return an extractive summary and utilized ROUGE scorer to record 80% precision.

House Price Predication Using ML.

May 2023

- Engineered Python algorithms to predict house prices in Bangalore with accuracy of 85%.
- Built SQL database for efficient data management and user-friendly front-end, conducted data analysis ensuring system scalability and reliability, resulting in a 20% increase in prediction accuracy.
- Built using HTML, CSS Python, MySQL, Pandas, NumPy.

Analyzing Online Shoppers' Intention

November 2020

- Led a team of 5 in conducting exploratory analysis on a dataset of 12500 online customers, helping improve performance of e-commerce websites.
- Forecasted demands based on visualizations made using NumPy, Pandas, Matplotlib.

Library Management

December 2019

- Designed a website for managing an online library system enabling librarians to simplify tasks and enhance productivity.
- Deployed website using HTML, CSS, Bootstrap for UI, and phpMyAdmin and MySQL for databases.

LEADERSHIP AND INVOLVEMENT

- **Awards:** Ranked first among 225 students in the Computer Department in academic years 2019-2023.
- **Member:** Led Social Responsibility Team during academic years 2020. Collectively organized events such as blood donation drives, flood relief camps, traffic monitoring and spread awareness about NGOs across Bangalore.

CERTIFICATIONS & LICENSES

- AWS Certified Cloud Practitioner Amazon Web Services (AWS) • Issued Apr 2025
- AWS Certified Machine Learning - Specialty Amazon Web Services (AWS) • Issued Apr 2025
- AWS Certified AI Practitioner Amazon Web Services (AWS) • 608e777a30a744f4b73c1282d9d7c836 • Issued Apr 2025